

# Social Media Engagement Analytics

## Data Visualization & Analysis Report

### 1. Introduction

This report analyses social media data to identify patterns in content performance, user behaviour, and engagement trends. Various visualization techniques and statistical summaries were used to derive insights from the dataset.

### 2. Content Performance Analysis

- Video posts show the highest engagement rate (1.12), followed by text posts
- Food category performs best (1.35), while travel has the lowest engagement
- Brazil and Australia have the highest average engagement rates
- USA and India show comparatively lower engagement

### 3. User Trends

- No significant relationship between age and engagement (correlation  $\approx 0.008$ )
- Engagement is evenly distributed across different age groups
- Verified accounts have slightly higher engagement than non-verified users
- Verification status provides a moderate advantage

### 4. Behavioural Insights

- No clear best time of day identified due to limited hourly data
- Mobile devices have the highest average watch time
- Desktop and tablet users show similar but slightly lower watch time
- Device type has a minor influence on engagement behaviour

### 5. Sentiment Analysis

- Negative sentiment posts have the highest engagement (1.03)
- Neutral sentiment shows moderate engagement
- Positive sentiment has the lowest engagement
- Negative content tends to drive more interaction

### 6. Overall Conclusion

The analysis shows that engagement is influenced more by **content characteristics** than by user demographics. Post type and category play a key role, while factors like age have minimal impact. Behavioural patterns such as device usage and sentiment also contribute to engagement, but to a lesser extent.